PROJECT BRIEF

Autonomous Data Science Co-Pilot

Summer Internship 2025

Project Type	Agentic AI / Data Engineering / Full-Stack Python

1. Problem Statement

Business teams sit on mountains of raw data — CSVs, Excel sheets, JSON exports — but lack the expertise to extract insights. This project challenges you to build an Autonomous Data Science Co-Pilot: an AI-powered agent that acts like a junior data analyst. A non-technical user uploads a file, asks a question in plain English, and the agent independently writes Python/Pandas code, executes it in a sandboxed environment, self-corrects errors using RAG over official documentation, and delivers professional charts and insights — zero coding required from the user.

2. Core Concept & Key Innovation

- User uploads a data file (CSV / Excel / JSON) and asks a question in plain English.
- The AI Agent writes the necessary Python / Pandas code autonomously.
- Code is executed in a background sandbox; output is captured.
- On error, the agent uses RAG over official Python & Pandas manuals to self-correct — it does not give up.
- Corrected code re-runs until a clean chart or insight is delivered.

Key Differentiator: Moves from "Text-Out" to "Action-Out" — delivers a finished artefact, not just instructions.

3. Use Cases

#	Use Case	Description
1	Sales Dashboard	Upload a monthly sales CSV — get an instant revenue-by-region bar chart.
2	Data Quality Audit	Detect missing values, outliers, and duplicates; produce a cleanliness report.
3	Trend Analysis	Ask 'Is my traffic growing?' over a time-series CSV and receive a trend line with key observations.
4	Cohort Analysis	Auto-segment customers by spend or activity and produce grouped visualisations.
5	Ad-hoc Queries	Founders query operational data without writing a single line of code.

4. Recommended Tech Stack

Layer	Tools / Libraries
Frontend / UI	Streamlit or Gradio
LLM / Agent	OpenAI GPT-4o or open-source LLM via LangChain / LlamaIndex
Code Execution	Python subprocess sandbox (restricted exec environment)
RAG Pipeline	FAISS or ChromaDB + documentation corpus
Data Processing	Pandas, NumPy, OpenPyXL, json

Layer	Tools / Libraries
Visualisation	Matplotlib, Seaborn, Plotly

5. Prerequisites for Interns

5.1 Must-Have (Before Day 1)

- **Python:** Python 3.10+, functions, file handling, error handling, virtual environments.
- **Data Libraries:** Pandas & NumPy — CSV reading, filtering, groupby, basic data manipulation.
- **Visualisation:** Matplotlib or Seaborn for basic charts.
- **Version Control:** Git — clone, commit, push, pull requests.

5.2 Good to Have

- **LLM API:** Any LLM API — OpenAI, Gemini, Anthropic, or Hugging Face.
- **Prompting:** Basic prompt engineering — system prompts, few-shot examples.
- **Web App:** Built a Streamlit or Flask app before.
- **Vector DBs:** Conceptual understanding of embeddings and vector similarity.

5.3 Will Be Taught During Internship

- LangChain / LlamaIndex agent orchestration
- RAG architecture — chunking, embedding, retrieval, re-ranking
- Sandboxed code execution and self-correction loop design
- FAISS / ChromaDB setup and querying

6. Expected Deliverables

- Web-based UI (Streamlit) — upload data, ask questions in plain English, get instant charts.
- Autonomous agent that writes, executes, and self-corrects Python/Pandas code.
- RAG pipeline over official Python and Pandas documentation.
- Support for CSV, Excel (.xlsx), and JSON file formats.
- GitHub repository with clean code, README, and setup instructions.
- Final demo covering at least 5 end-to-end use case demonstrations.

Good luck, and welcome to the team!